

$$b_n'' - \alpha_n b_n = c_n$$

$$\bullet \quad \tilde{b}_n = c_n - \alpha_n b_n$$

$$\tilde{b}_n'' = b_n$$

$$e^{\pm n\pi} \quad [\text{Answer}]$$

$$(i) \quad e^{\pm i n \pi} = \pm 1 = \cos(n\pi) + i \sin(n\pi)$$

$$\pm 1$$

$$(ii) \quad e^{\pm n\pi} \left[A_n \cosh(\cdot) + B_n \sinh(\cdot) - \frac{c_n}{(\cdot)} \right]$$

$$= \hat{A}_n \cosh(\cdot) + \hat{B}_n \sinh(\cdot) - \frac{\hat{c}_n}{(\cdot)}$$

$$A_n$$

* Inner product: $\langle \cdot, \cdot \rangle$ $V = \mathbb{R}^3 : (x, y, z)$

Two vectors $\uparrow$

ex: "Dot-product": $\vec{x} \cdot \vec{y} = x_1 y_1 + x_2 y_2 + x_3 y_3$

also: $\langle \vec{x}, \vec{y} \rangle \leftarrow \text{(Notation)}$

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

V: Vector space

* Conditions for defining this

ex: $\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$

$\hookrightarrow (\vec{x} + \vec{y}) \cdot \vec{z} = \vec{x} \cdot \vec{z} + \vec{y} \cdot \vec{z}$

S.L.T: ODE: $\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x)y = -\lambda w(x)y$

"standard form"

~ Idea: transform other ODE's into this standard form

(1a) $y'' + 3y' + y = -\lambda y$

$\hookrightarrow$ sub ex: $p(x) = 1 : y'' + qy = -\lambda w(x)y$

$p(x) = x : \underbrace{\frac{d}{dx}(x)}_1 y' + x \frac{d}{dx}(y') + q(x)y = -\lambda w(x)y$

$y' + x y'' + q(x)y = \uparrow$

* General Case: *

$$F(x) y'' + Q(x) y' + R(x) y = -\lambda V(x) y$$

$F(x) \neq 0$

$$y'' + \frac{Q}{F} y' + \frac{R}{F} y = -\lambda \frac{V}{F} y$$

int. factor: $r(x) = \exp\left(\int \frac{Q(x)}{F(x)} dx\right)$

$$* r y'' + r \frac{Q}{F} y' + \frac{R}{F} y = -\lambda \frac{Vr}{F} y$$

$$L_r = \frac{d}{dx}(r(x)) = \frac{d}{dx}\left(e^{\int \frac{Q}{F} dx}\right)$$

$$= e^{\int \frac{Q}{F} dx} \frac{d}{dx}\left(\int \frac{Q}{F} dx\right)$$

$$= e^{\int \frac{Q}{F} dx} \frac{Q}{F}$$

$$= r \frac{Q}{F}$$

$$\hookrightarrow r y'' + r' y' + \frac{R}{F} y = -\lambda \frac{Vr}{F} y$$

$$\frac{d}{dx}(r y') + \frac{R}{F} y = -\lambda \frac{Vr}{F} y$$

$$P y'' + Q y' + R y = -\lambda V y$$

$$p(x) = r(x) = \exp\left[\int \frac{Q}{F} dx\right]$$

$$q(x) = R(x)/F(x)$$

$$w(x) = V/F$$

$$(1a) y'' + 3y' + y = -\lambda y$$

$$F=1, Q=3, R=1, V=1$$

$$(1b) x^2 y'' - x y' + y = -\lambda y$$

$$F=x^2, Q=-x, R=1, V=1$$

$$(1c) (x^2+1) y'' + x y' = -\lambda y$$

$$F=x^2+1, Q=x, R=0, V=1$$

ex: All previous cases that we had before were examples of S.L.T

Recall: $u(x,t) = X(x) T(t) \rightarrow$ PDE $\rightarrow$ Two ODEs

$X(x)$:

$$X'' = -\lambda X$$

X_n : Eigenfunctions

λ_n : Eigenvalues

S.L.: $F = 1$, $Q, R = 0$, $V = 1$

* $X(0) = X(L) = 0$, $0 < x < L$

* $X_n = A_n \sin(\frac{n\pi x}{L})$, $\lambda_n = \frac{n^2 \pi^2}{L^2}$

↳ Eigenfn: Inner product:

$$\langle X_n, X_m \rangle = \int X_n(x) X_m(x) dx$$

$$V = \{X_m\}_{m=0}^{\infty}$$

* Orthonormal basis elements:

Defined:

$$\langle \tilde{X}_m, \tilde{X}_n \rangle = \delta_{nm}$$

• Obtain X_n : want $\tilde{X}_m$

↳ ex: $X_n = A_n \sin(\frac{n\pi x}{L})$

To find A_n , plug into $\langle \cdot, \cdot \rangle$ and want this to be equal to δ_{nm}

$$\begin{aligned} \langle X_n, X_m \rangle &= \int_0^L A_n \sin(\frac{n\pi}{L}x) A_m \sin(\frac{m\pi}{L}x) dx \\ &= A_n A_m \int_0^L \sin(\frac{n\pi}{L}x) \sin(\frac{m\pi}{L}x) dx \\ &= A_n A_m \frac{L}{2} \delta_{nm} \\ &= \delta_{nm} \end{aligned}$$

* If $n \neq m$ $\delta_{nm} = 0$

if $n = m$: $A_n A_n \frac{L}{2} = 1$

$$A_n^2 = \frac{2}{L}$$

$$A_n = \sqrt{\frac{2}{L}}$$

$$\tilde{X}_n = \sqrt{\frac{2}{L}} \sin(\frac{n\pi x}{L})$$

(2)

(b) $\langle A_n X_n, A_m X_m \rangle = \delta_{nm}$

(3) Similar to the example, different B.C.

$\leadsto$ Cases for $\lambda = 0$

$\lambda \neq 0$ etc...

(4) Idea: PDE $\rightarrow X(x)T(t) \rightarrow$ ODE $X(x)$
use S.L.T.

